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Impact of Temporal Order Selection on Clustering Intensive Longitudinal Data Based on Vector Autoregressive Models

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Abstract
When multivariate intensive longitudinal data are collected from a sample of individuals, the model-based clustering (e.g., vector autoregressive [VAR] based) approach can be used to cluster the individuals based on the (dis)similarity of their person-specific dynamics of the studied processes. To implement such clustering procedures, one needs to set the temporal order to be identical for all individuals; however, between-individual differences on temporal order have been evident for psychological and behavioral processes. One existing method is to apply the most complex structure or the highest order (HO) for all processes, while the other is to use the most parsimonious structure or the lowest order (LO). Up to date, the impact of these methods has not been well studied. In our simulation study, we examined the performance of HO and LO in conjunction with Gaussian mixture model (GMM) and *k*-means algorithms when a two-step VAR-based clustering procedure is implemented across various data conditions. We found that (a) the LO outperformed the HO in cluster identification, (b) the HO was more favorable than the LO in estimation of cluster-specific dynamics, (c) the GMM generally outperformed the *k*-means, and (d) the LO in conjunction with the GMM produced the best cluster identification outcome. We demonstrated the uses of the VAR-based clustering technique using the data collected from the "How Nuts are the Dutch" project. We then discussed the results from all our analyses, limitations of our study, and direction for future research, and meanwhile offered our recommendations on the empirical uses of the model-based clustering techniques.

Translational Abstract
When multivariate intensive longitudinal data are collected from a sample of individuals, the model-based clustering (e.g., vector autoregressive [VAR] based) approach can be used to cluster the individuals based on the (dis)similarity of their person-specific dynamics of the studied processes. To implement such clustering procedures, one needs to set the temporal order to be identical for all individuals; however, between-individual differences on temporal order have been evident for psychological and behavioral processes. One existing method is to apply the most complex structure or the highest order for all processes, while the other is to use the most parsimonious structure or the lowest order. Up to date, the impact of these methods has not been well studied. In our simulation study, we examined the performance of highest order and lowest order in conjunction with Gaussian mixture model and *k*-means algorithms when a two-step VAR-based clustering procedure is implemented across various data conditions. We also demonstrated the uses of the VAR-based clustering technique using the data collected from the "How Nuts are the Dutch" project. We then discussed the results from all our analyses, limitations of our study, and direction for future research, and meanwhile offered our recommendations on the empirical uses of the model-based clustering techniques.

Keywords: intensive longitudinal data, clustering, model-based clustering, Gaussian mixture model, *k*-means

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Intensive longitudinal data (ILD) provide researchers a unique opportunity to uncover dynamical mechanism underlying time-ordered observations of psychological or behavioral processes, such as emotion (e.g., Pe & Kuppens, 2012), affect (e.g., Jahng et al., 2008), stress (e.g., daSilva et al., 2021), interpersonal behaviors (e.g., Roche et al., 2014), and psychosomatic characteristics (e.g., Price et al., 2017; Von Leupoldt et al., 2006). In real-world research, ILDs are often collected from a group of individuals of interest, such as depression suffers (e.g., Axelson et al., 2003; Whalen et al., 2007), which enable researchers to model not only the within-individual dynamics of the studied processes but also the between-individual differences on the within-individual dynamics.

A variety of statistical techniques have been proposed to analyze the ILDs of multiple individuals. Depending on the ways of modeling between-individual differences or heterogeneity, those techniques can be categorized into two classes. One class of the techniques quantifies such differences as quantitative. For example, one can employ mixed-effect modeling techniques to summarize between-individual differences on dynamics by assuming that such differences follow certain probability distributions. Here, the within-individual dynamics can be modeled using the techniques originated from social sciences, such as dynamic factor models (e.g., H. Song & Ferrer, 2012), as well as those that have been applied extensively in engineering and econometrics, such as state-space models (e.g., Chow et al., 2007) and time series models (e.g., Babbin et al., 2015; Bringmann et al., 2013; Liu et al., 2013; Stefanovic et al., 2022). In addition, between-individual differences on dynamics can also be modeled by using multilevel structural equation modeling (SEM) approaches such as dynamic SEM (e.g., Asparouhov et al., 2018; Hamaker et al., 2018; D. McNeish & Hamaker, 2020; H. Song & Zhang, 2014).

In contrast with the quantitative methods, clustering is another class of techniques that quantifies between-individual differences in dynamics as qualitative and assumes that such differences reflect distinct types of dynamic processes among individuals. The goal is to assign individuals into different clusters where the similarity in dynamic features or model coefficients are maximized within clusters but minimized between clusters. The cluster-wise approach has long been used as a data mining or data reduction tool in engineering, finance, health, and neuroscience (e.g., Aghabozorgi & Teh, 2014; Elangasinghe et al., 2014; Stetco et al., 2013; Tran & Wagner, 2002; Yan et al., 2022). However, it has not received much attention in social and behavioral research. Meanwhile, large volumes of ILDs have become much easier to collect because of the recent advancement in data collection techniques, such as ecological momentary assessment (Shiffman et al., 2008), which provide researchers with valuable opportunities to examine the patterns of various psychological and behavioral processes. For example, clustering of ILDs can help to identify distinct groups of smokers who share similar dynamics of short- or long-term swings on mood, affect, and smoking cravings during cessation periods, by which group-specific prevention and intervention could be developed and implemented to achieve effective outcomes.

Model-based clustering is one of the techniques commonly used for clustering ILDs. It typically starts with fitting a specific model (e.g., time series model, state-space model, or unified SEM) to each individual ILD and then feeds the estimated coefficients to clustering algorithms for cluster identification (e.g., Gates et al., 2010; Kim et al., 2007). People who are placed into the same clusters are assumed to share similar dynamic features for the behavioral system under study while differentiating themselves from those in other clusters. After the clusters and their members are identified, one can estimate the cluster-specific dynamics to determine the unique dynamic nature of each identified cluster. In addition, between-individual differences can be examined at the cluster level through linking the cluster membership with potential correlates and distal outcomes.

One special requirement by the current model-based clustering procedures is that the number of model parameters supplied to the clustering algorithms needs to be the same across all individual ILDs, which imposes the temporal order to be identical for all individuals in the sample. However, interindividual differences on temporal orders have been widely reported for various psychological or behavioral processes (e.g., Bosley et al., 2020; Chow et al., 2007; Gu et al., 2014; Rosmalen et al., 2012; Snippe et al., 2015). For example, Ernst et al. (2021) reported that the ILDs of daily affect can be best described by a Lag 1 process for 94% of the 366 participants but by Lag 2 or Lag 3 models for the remaining 6%. Similarly, Snippe et al. (2015) found that the lagged relationship between repetitive thinking and depressive symptoms varied substantially across participants, ranging from no effect to Lag 1 effect and to Lag 2 effect.

When temporal orders differ across individuals, as is very likely to occur, selecting an appropriate order for all individuals becomes an important issue to address before applying any model-based clustering procedures. Two methods have been proposed for temporal order selection. One applies the most complex structure or highest order (HO) for all individual processes (e.g., Ernst et al., 2020; Kalpakis et al., 2001; Maharaj, 2000; Ren & Barnett, 2022; Xiong & Yeung, 2004), while the other chooses the most parsimonious structure or the lowest order (LO) for all individuals (e.g., Bulteel et al., 2016; Ernst et al., 2021; Gates et al., 2017; Zheng et al., 2013). As a result, the utilization of the HO likely leads to overfitting, whereas the LO practice likely leads to underfitting, at least for some of the individual ILDs in the data. How each method of temporal order selection, HO versus LO, would perform on cluster identification as well as estimation of cluster-level dynamics? In other words, what are the consequences of overfitting and underfitting when clustering individual processes with the model-based approach? To our best knowledge, these questions have not been well answered yet, leaving the common practices of the HO and the LO, as well as the model-based clustering techniques, unwarranted for social and behavioral research.

The goal of this study was three-fold. First, we introduced the model-based approach for clustering multiple ILDs to behavioral researchers. Second, we examined the effect of temporal order selection on ILD clustering by comparing the performance of HO and LO using simulated data. Third, we illustrated the use of the model-based clustering technique through an application with empirical data. Overall, this study not only deepened our understanding of the model-based clustering techniques in data and behavioral sciences but also provided guidance to applied researchers in fields such as health and psychopathology, which would enable them to apply the intensive longitudinal paradigm with an improved accuracy in clustering behavioral systems of interests and thus designing more targeted, effective intervention or treatment.

In the following, we first introduced the model-based approach for clustering ILDs. We then described the vector autoregressive (VAR) model (Hamilton, 1994; Lütkepohl, 2015), the one used as the "model" in the model-based approach in this study. Next, we presented a two-step procedure for clustering multiple ILDs based on VAR models, with an emphasis on two commonly used clustering algorithms. We then reported our simulation study, where the effect of temporal order selection techniques was examined comprehensively. We provided an empirical analysis to further illustrate the uses of HO and LO in conjunction with different clustering algorithms. Finally, we discussed our findings and offered recommendations on the uses of the model-based clustering techniques.

Model-Based Approach for Clustering ILDs

ILDs can be clustered using three different approaches based on shapes, features, or generative models of the observed time series (see Aghabozorgi et al., 2015; Liao, 2005). The shape-based approach clusters raw time series through directly leveraging the time series-specific similarity or distance (e.g., Euclidean distance) in conjunction with standard algorithms (e.g., the *k*-means). Although it is simple to implement, this approach is subject to shortcomings of lacking an effective way to deal with multiple ILDs with varying lengths, missing values, and/or unequal intervals.

Unlike the shape-based approach, the feature-based approach typically involves two stages in the clustering process. It first extracts a set of features from each ILD (e.g., mean, autocorrelations, and independent components) and then employs standard algorithms in the feature space to cluster multiple ILDs. The advantage of this approach is that classic algorithms like the *k*-means can be performed, which waives the requirement of developing novel algorithms for clustering the extracted features. However, crucial information contained in the original data could be lost in the feature extraction stage, which could lead to poor clustering outcomes.

The model-based approach clusters ILDs based on the estimated coefficients of the chosen model assumed to generate the observed ILDs. In comparison with the other two approaches, this approach could be more advantageous in behavioral research because (a) it works in a reduced dimensional space and therefore clusters multivariate time series with a higher efficiency and (b) it estimates the dynamical coefficients within and across variables, such as autoregressive (AR) and cross-regressive (CR) coefficients numerically characteristic of the dynamics of human behavioral systems. The model-based approach can be implemented in different ways, depending on the number of steps involved in conjunction with specific algorithms used for clustering. In a one-step procedure, model estimation and cluster formation are performed iteratively. For example, a cluster-specific VAR(1) model could be estimated for each cluster in which the initial memberships are assigned randomly. Then, the differences between each person's data and their cluster-specific VAR(1) estimates are calculated as prediction errors. Individuals are reassigned to the clusters iteratively until the prediction errors get minimized and the cluster memberships remain constant (e.g., Bulteel et al., 2016).

In contrast, a two-step procedure runs the model estimation and clustering processes as two separate steps. For example, Zheng et al. (2013) fitted a VAR(1) model to each individual in Step 1 and then calculated the similarities (i.e., Euclidean distance) between each pair of individuals based on their person-specific coefficients, through which clusters were identified in Step 2 by using a hierarchical clustering algorithm. In another example, Ernst et al. (2021) submitted the person-specific model coefficients estimated in Step 1 to the finite Gaussian mixture model (GMM) for cluster identification in Step 2, under the assumption that the model coefficients follow a mixture of normal distributions. It should be noted that the two-step approach treats the estimated coefficients from Step 1 as true values in Step 2. Whether or how the potential bias produced at Step 1 would impact the clustering outcome at Step 2 remains unknown.

The major drawback of the one-step method lies in the fact that it does not allow within-cluster variation on dynamical coefficients, but the two-step procedure does. In addition, the one-step clustering method partitions individuals into distinct clusters through minimizing within-cluster predictive errors, whereas the two-step method forms clusters by quantifying the (dis)similarity of estimated dynamical coefficients. Research has shown that the two-step approach outperformed the one-step approach in clustering multivariate ILDs over a range of conditions (e.g., Ernst et al., 2021). In this study on temporal order selection, we chose a two-step procedure for clustering multivariate ILDs in both the simulation and empirical analyses.

Analysis of Single-Individual ILDs With VAR Models

Time series models have been widely used in various fields to model dynamical mechanisms underlying time-ordered observations from single individuals or units. When a dynamical system consists of multivariate processes, multivariate time series models are needed. Among them, VAR models have recently gained popularity in social and behavioral research because of their capacity of modeling temporal dependence within and between variables through AR and CR components. For example, Rosmalen et al. (2012) used this technique to investigate the temporal relationship between depression and physical activity, and they found that patients' current depression level can be predicted by their physical activity at previous occasions.

A VAR(p) model indicates that the current state of a dynamic process is predicted by its state in p immediately preceding occasions, where p represents the number of lags or temporal order of the process. For example, a bivariate VAR(2) model can be illustrated by the following:

$$Y_t = A + \Phi_{t-1}Y_{t-1} + \Phi_{t-2}Y_{t-2} + E_t. \tag{1}$$

or

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}_t = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix}_{t-1} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}_{t-1} + \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix}_{t-2} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}_{t-2} + \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \end{bmatrix}_t. \tag{2}$$

where Y_t is a vector of observed variables at time t, the transition matrices Φ_{t-1} and Φ_{t-2} contain the AR coefficients in the diagonals and the CR coefficients in the off diagonals, E_t represents a vector of residuals or innovations, and A is the intercept vector.

Overall, the AR and CR coefficients represent how quickly the process returns to the mean, while the innovation component represents a collection of internal or external events that pushes the process away from its mean at a given time. The temporal dependencies captured by the AR and CR coefficients are assumed not too strong, so the perturbations because of the innovation "die out" over time rather than accumulating (Hamilton, 1994). It should be noted that, like most of the time series models, the VAR model requires the time series to be stationary; that is, the statistical properties of the data (e.g., mean, variance, and autocovariance) need to be invariant over time.

A critical step in fitting a VAR(p) model to a single-individual ILD is to determine the temporal order p. Information criteria such as Akaike information criterion (AIC) or Bayesian information criterion (BIC) can aid this selection after running a series of VAR models with different p values; however, incorrect decisions can still be made and as a result, overfitting/underfitting could occur.[1] When the selected p is greater than the true order of the studied processes, overfitting occurs; when it is smaller than the true order, underfitting occurs. Research has shown that overfitting and underfitting are associated with different bias when fitting time series models to individual ILDs. For example, Lütkepohl (2015) showed that overfitting of a VAR model causes increased prediction errors, while underfitting often generates autocorrelated errors. Furthermore, including additional lags increases the number of estimated parameters, which would require many measurement occasions for reliable parameter estimation. This poses challenges for typical behavioral research where ILDs are collected from a relatively small number of time points. Overall, an incorrect specification of p would lead to biased estimation at the individual level (Braun & Mittnik, 1993). How such biased estimation could impact the clustering accuracy needs to be addressed.

Clustering Multiple-Individual ILDs Based on VAR Models

When a targeted dynamical system consists of multidimensional facets, the realized multivariate ILDs can be clustered based on VAR models. As mentioned previously, we chose to use a two-step clustering procedure for clustering multivariate ILDs. In Step 1, a VAR(p) model is fitted to each individual ILD, by which person-specific AR and CR coefficients up to lag p are estimated; in Step 2, a specific clustering algorithm is performed on the estimated coefficients to cluster individuals into different groups. The performance of this two-step procedure primarily depends on the choices of p in Step 1 as well as the specific algorithms in Step 2. While our major task was to examine the effects of HO and LO regarding the choices of p, we also investigated whether and how those effects depend on the specific algorithms chosen in Step 2.

Clustering algorithms can be classified as probabilistic and nonprobabilistic algorithms. Probabilistic algorithms impose certain probability distributions, such as a mixture of normal distributions over the parameters of a chosen model (e.g., AR and CR coefficients in a VAR model) and then identify the clusters of individual ILDs whose coefficients follow the same distribution (Fraley & Raftery, 2002; McLachlan & Basford, 1988). Conversely, nonprobabilistic algorithms typically rely on distance or similarity metrics, such as Euclidean distance to partition individual ILDs into different clusters, by which the within-cluster similarities are maximized, and the between-cluster similarities are minimized.

In this study, we chose the finite GMM and the k-means algorithms as the representatives of the probabilistic and nonprobabilistic algorithms, respectively. The GMM algorithm (Banfield & Raftery, 1993; Browne & McNicholas, 2014; Celeux & Govaert, 1995) assumes that the AR and CR coefficients of a VAR(p) model come from a mixture of multivariate normal distributions, and each unique distribution corresponds to a distinct cluster. On the other hand, the k-means algorithm (MacQueen, 1967), as one of the most widely used clustering algorithms, was designed to partition ILDs into k clusters based on the squared Euclidean distance of raw data or extracted features. The GMM algorithm generates a probabilistic membership for each individual and then assigns individuals to a specific cluster based on the highest posterior probability. In contrast, the k-means algorithm assigns individuals to one and only one cluster with a probability of either 0 or 1. Overall, the GMM tends to provide greater flexibility in finding the true, underlying cluster structure present in the data (Steinley & Brusco, 2011).

It should be noted that high dimensionality, referring to the time length and/or the number of variables in a dynamical system, would negatively influence GMM's performance. High dimensionality could cause the feature space to have random dimensions that do not contribute to subsequent cluster identification (Scrucca et al., 2016). In addition, the goodness of cluster recovery with the GMM cannot be guaranteed because of potential overfitting occurred with exceedingly complex models (Steinley & Brusco, 2011). In contrast, when high dimensionality becomes a concern, the k-means clustering could avoid model overfitting and consequently produce a parsimonious cluster structure.

When the two-step procedure with a VAR(p) is used to cluster a sample of multivariate ILDs, overfitting would occur on those individual processes with true orders lower than the chosen p. This likely yields some model coefficients especially at higher temporal orders not varying across the true clusters. Such coefficients are not informative for cluster identification and therefore labeled as noneffective coefficients. They act more like noises in the clustering stage. Clustering accuracy could be attenuated if the procedure produces many noneffective coefficients (Takano et al., 2020). In addition, as p increases, the number of estimated coefficients of a VAR model will increase. As a result, the feature space would become increasingly sparse, which will blur the similarities among clusters (Steinbach et al., 2004). In this case, the clustering algorithms that partition data based on similarities would perform poorly. Research has shown that time series models with high temporal orders severely restricted clustering accuracy (Bouveyron & Brunet-Saumard, 2014; Scott & Thompson, 1983). Regarding underfitted VAR models, the unmodeled temporal relationships could manifest themselves as autocorrelated residuals that lead to biased estimation of the model coefficients, which may in turn impact the accuracy of clustering.

Previous studies on clustering ILDs mostly used Lag 1 time series models, such as a VAR(1) model (e.g., Bulteel et al., 2016; Ernst et al., 2021). This practice tends to cause underfitting than overfitting because higher lag orders may exist for some individual ILDs but left unmodeled. As a matter of fact, current knowledge is very limited about the impact of temporal order selection, or overfitting/underfitting on clustering multivariate ILDs using the model-based approach. Thus, we conducted a simulation study and an empirical analysis to offer some insights on this issue, thereby providing guidance on the applications of the relevant techniques.

A Simulation Study

The purpose of the simulation study was to investigate the impact of temporal order selection (HO vs. LO) on clustering multivariate ILDs under a broad range of conditions. In real-world scenarios, multivariate ILDs collected from a sample of participants rarely follow identical dynamic processes. For instance, some may follow a

[1] Research showed that AIC and BIC tend to underestimate the true lag orders in shorter ILDs (Kilian, 2001).

VAR(1) process, while others may follow a VAR(2) process. In the cases where both processes exit in the data, what would happen if all individuals were clustered based on a VAR(1) (i.e., the LO strategy), under which underfitting would occur for the ILDs with a generative model of VAR(2), or if all individuals were clustered based on a VAR(2) (i.e., the HO strategy), under which overfitting would occur for the ILDs with a generative model of VAR(1)? This is the key question we addressed with the simulation study. Additionally, we explored whether the impact of the HO or LO on clustering depends on the specific algorithms used to cluster individuals. We evaluated three specific outcomes of the clustering procedures under each condition, including the percentage of correctly identified number of clusters, the recovery of cluster membership, and the accuracy of estimated cluster-specific coefficients. The code for this simulation has been made publicly available at https://github.com/ild-clustering/temporal_order_selection.

Data Generation

Data were generated using both VAR(1) and VAR(2) models. Six specific factors were manipulated: (a) the proportion of the data generated by VAR(2): 0%, 10%, 40%, 60%, 90%, and 100%; (b) the length of the generated time series: 50, 100, 500, and 1,000; (c) the number of individuals in the sample: 60, 120, and 240; (d) the number of Clusters: 2, 3, and 4; (d) the Euclidean distance between clusters: 0.10, 0.15, 0.20, and 0.25; and (e) the proportion of effective coefficients within cluster: 25%, 37.5%, 50%, 62.5%, and 75%. Given the VAR-based clustering procedure is computationally intensive, we generated only 10 replicates per condition in our simulations. This number aligned well with those used in previous simulation studies involving clustering (e.g., Bulteel et al., 2016; Ernst et al., 2021; Steinley & Brusco, 2011). In total, we generated 34,560 unique data sets by fully crossing all the factors. The within-data set factors were the temporal order selection method (HO vs. LO) and the clustering algorithms (i.e., the GMM and *k*-means algorithms).

Individual multivariate ILDs were generated using the VAR(p) model shown in Equation 2. The number of variables was chosen to be 4, the temporal order p was set to be either 1 or 2 depending on the predetermined percentages of each process, and the intercepts were set to zero for simplicity. The distance between different clusters was reflected through the distance between cluster-specific AR and CR coefficients in the Φ matrix of Equation 2. The same VAR models were used to generate ILDs for all individuals in the same clusters. That is, the Φ matrix at the same lag was set to be identical for individual ILDs within cluster.[2] The within-cluster variation came from innovation terms E_t, which were sampled from a multivariate normal distribution $(0, I + 0.5)$.

We first generated a reference cluster with the AR and CR coefficients randomly sampled from uniform distributions. Specifically, the AR coefficients in Lag 1 were sampled from a uniform distribution of [0.5, 0.7], and the CR coefficients in Lag 1 followed a uniform distribution of [−0.4, 0.4]. The two distributions were purposely chosen to sample the AR and CR coefficients at a moderate level for the reference cluster, which allowed for generating small or large coefficients for other clusters, while stationarity can be ensured. If a Lag 2 process was generated, both AR and CR coefficients at Lag 2 were randomly generated from a uniform distribution of [−0.2, 0.2], resulting in smaller coefficients than their counterparts at Lag 1. By using the aforementioned uniform distributions, the possible values generated for the AR and CR coefficients at Lag 1 and Lag 2 aligned well with those used in prior simulation studies (Bulteel et al., 2016; Ernst et al., 2021, 2020; Lafit et al., 2022) as well as those obtained from the empirical applications using the VAR(p) models (e.g., Krone et al., 2018; Rosmalen et al., 2012). After the reference cluster was generated, the AR and CR coefficients for other clusters were generated by adding or subtracting the predetermined values of distance from those coefficients in the reference cluster. The effective coefficients were randomly allocated in the Φ matrix. If a Lag 2 process was generated, the effective coefficients were evenly located in both Lag 1 and Lag 2 in the Φ matrix. To ensure stationarity, the data generation procedures mentioned above were repeated until all cluster-specific VAR processes met the assumption of stationarity; that is, the eigenvalues of the Φ matrix are less than 1 in modulus.

In each simulated data set, the size of each cluster was determined by the proportion of ILDs generated by the VAR(2) model. Specifically, when the proportions were 0% or 100%, the data were generated to be evenly distributed across clusters. When the proportions were 10% or 90%, one cluster contained 10% of the data, while the remaining ILDs were evenly distributed across the other clusters. When the proportions were 40% or 60%, one cluster contained 40% of the ILDs and the rest were evenly distributed across the remaining clusters. For instance, in a simulated data set with 10% of VAR(2) processes and four clusters, one cluster contained 10% of the ILDs that followed the VAR(2) process, whereas the remaining 90% were evenly distributed across the other three clusters, with each cluster containing 30% of these ILDs who followed the same cluster-specific VAR(1) process.

Clustering Procedure

In each simulated data set, the person-specific AR and CR coefficients were first estimated for every single individual using both LO and HO methods. For example, in a data set with 10% VAR(2) and 90% VAR(1), we fitted a VAR(1) to each individual's ILDs under the LO method and a VAR(2) to each individual's ILDs under the HO method. Once the VAR model was estimated, the GMM and *k*-means algorithms were then applied to the person-specific AR and CR coefficients for cluster identification, which resulted in a total of 16 coefficients per person under the LO and 32 coefficients per person under the HO to be clustered. We requested the number of clusters to be one to five under both algorithms. The GMM algorithm utilized BIC to determine the optimal number of clusters as well as the optimal cluster covariance structures. The *k*-means algorithm used the gap statistic (Tibshirani et al., 2001) to identify the optimal number of clusters. To ensure the stability and accuracy of the gap statistics, 500 bootstrapped samples were used for each *k*-means clustering. In an effort to minimize the potential effect of random starts on the clustering outcomes, we used 10 random starts

[2] Research showed that increasing differences in the AR and CR coefficients between individuals within cluster decrease the performance of model-based clustering (Ernst et al., 2021). In our simulations, we purposefully set the AR and CR coefficients to be identical across individuals, which allowed for a more focused investigation of the effect of temporal order selection on the clustering outcomes (without contamination caused by the between-individual differences on the Φ matrix).

and 1 rational start for the GMM algorithm, and 5,000 random starts and 1 rational start for the k-means algorithm, with the rational starts being derived from hierarchical clustering (e.g., Ernst et al., 2021; Fraley & Raftery, 1998; Steinley & Brusco, 2007, 2011).

Results

The recovery on the number of clusters was assessed by calculating the percentage of the simulated data sets where the number of clusters was correctly identified under a specific condition. Overall, the LO (72.9%) demonstrated a superior performance over the HO (49.5%), across all conditions. When the two algorithms were compared, the GMM showed higher average recovery rates (83.4% for the LO and 50.6% for the HO) than the k-means (62.3% for the LO and 48.5% for the HO) across all conditions.

Table 1 shows that regardless of temporal order selection methods or clustering algorithms used, higher recovery rates were associated with longer ILDs, greater distance between clusters, larger sample sizes, and larger number of effective coefficients. A closer examination further revealed that the increase in the sample size was associated with a larger increment on the recovery rates of the GMM than that of the k-means. However, the increase in the length of the time series was associated with a larger increment on the recovery rates of the k-means than that of the GMM. In other words, the GMM was more sensitive to the sample size, whereas the k-means was more sensitive to the length of the time series. A special note should be made for the short ILDs. The clustering accuracy was not satisfactory at $t = 50$, with the best average recovery observed under the LO in conjunction with the GMM (64.2%), and the worst observed under the HO in conjunction with the k-means (6.8%). In addition, $t = 100$ seemed to be a minimum requirement for achieving reasonable clustering accuracy (>70%) under the LO, but a much higher t would be needed to achieve this under the HO.

When the proportions of the VAR(2) processes varied from 0% to 100%, the mean accuracy of the clustering changed inconsistently. In the extreme case when all the ILDs in the sample followed a VAR(2), clustering based on a VAR(1) model still performed better than the clustering based on a VAR(2) model under both GMM (90% vs. 59.5%) and the k-means (78.4% vs. 65.2%). When all

Table 1
The Percentage of Correctly Recovered Number of Clusters

	Clustering algorithm							
	GMM				*k*-means			
	Temporal order selection methods							
	LO		HO		LO		HO	
Level	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
Proportion of VAR(2)								
0%	0.85	0.29	0.37	0.35	0.68	0.39	0.49	0.46
10%	0.84	0.26	0.5	0.37	0.47	0.43	0.33	0.42
40%	0.67	0.39	0.4	0.39	0.51	0.41	0.4	0.43
60%	0.91	0.2	0.62	0.38	0.77	0.31	0.63	0.43
90%	0.85	0.24	0.55	0.36	0.53	0.42	0.41	0.44
100%	0.90	0.22	0.6	0.39	0.78	0.32	0.65	0.43
Number of true clusters								
2	0.92	0.15	0.6	0.36	0.52	0.39	0.42	0.43
3	0.85	0.26	0.49	0.38	0.71	0.37	0.56	0.45
4	0.73	0.37	0.43	0.4	0.64	0.42	0.48	0.47
Sample size								
60	0.75	0.31	0.42	0.32	0.63	0.4	0.48	0.45
120	0.85	0.28	0.42	0.4	0.63	0.4	0.49	0.45
240	0.91	0.23	0.68	0.37	0.61	0.4	0.49	0.45
Length of ILDs								
50	0.64	0.34	0.25	0.27	0.29	0.31	0.07	0.16
100	0.81	0.28	0.33	0.32	0.53	0.37	0.28	0.35
500	0.94	0.19	0.65	0.36	0.81	0.34	0.76	0.38
1,000	0.94	0.19	0.79	0.31	0.86	0.3	0.83	0.34
Between-cluster distance								
0.1	0.74	0.34	0.37	0.36	0.47	0.42	0.36	0.44
0.15	0.83	0.28	0.48	0.38	0.58	0.41	0.45	0.46
0.2	0.88	0.25	0.56	0.38	0.68	0.37	0.52	0.45
0.25	0.9	0.22	0.61	0.38	0.76	0.34	0.6	0.43
Proportion of effective coefficients								
25%	0.77	0.33	0.38	0.35	0.53	0.42	0.4	0.44
37.50%	0.82	0.29	0.46	0.38	0.59	0.41	0.46	0.45
50%	0.85	0.28	0.52	0.39	0.63	0.4	0.49	0.45
62.50%	0.87	0.26	0.56	0.39	0.67	0.39	0.53	0.45
75%	0.88	0.25	0.6	0.39	0.69	0.38	0.55	0.45

Note. GMM = Gaussian mixture model; LO = lowest order; HO = highest order; VAR = vector autoregressive; ILD = intensive longitudinal data.

the ILDs followed a VAR(1), clustering based on a VAR(2) was associated with poor clustering outcome, 37.3% accuracy for the GMM and 48.5% for the *k*-means. When they were clustering based on a VAR(1), the accuracy on average was decent, 84.6% for the GMM, and 68.1% for the *k*-means. Putting it all together, underfitting was preferable to overfitting for cluster identification based on VAR models, although both produced a certain degree of clustering inaccuracy.

The recovery of cluster membership was evaluated by using the Adjusted Rand Index (ARI; Hubert & Arabie, 1985), a highly recommended method for measuring the degree of agreement between the predicted and the true cluster memberships (Milligan & Cooper, 1986; Steinley, 2004). ARI ranges from 0 to 1, where 0 indicates a random membership assignment and 1 indicates a perfect agreement between the predicted and the true memberships. According to Steinley (2004), ARI $\geq$ 0.90 is an excellent recovery, $\geq$0.80 means a good recovery, $\geq$0.65 indicates a moderate recovery, and $<$0.65 is a poor recovery.

Overall, clustering with the LO yielded a better average recovery of cluster membership (ARI = 0.80) than the HO (ARI = 0.55). Moreover, the GMM with the LO produced a good average recovery (ARI = 0.88), whereas the *k*-means with the LO yielded a moderate recovery (ARI = 0.73). On the other hand, when the HO was applied, both GMM and the *k*-means produced poor recovery with an average ARI of 0.56 for the former and 0.55 for the latter.

Table 2 reports the average ARIs for both clustering algorithms under all data conditions. Generally, higher ARIs were associated with longer ILDs, greater distance between clusters, larger sample size, and larger proportion of effective coefficients. This was consistent with the results on the recovery of the number of clusters. Notably, the GMM was much more sensitive to the changes of the sample size than the *k*-means did—the average ARIs increased from 0.80 to 0.94 under the LO and from 0.44 to 0.79 under the HO, whereas no apparent changes on the average ARIs were observed for the *k*-means under both the LO and the HO. Regarding the length of the ILDs, as t increased from 50 to 500, the cluster membership recovery improved from poor (ARI = 0.36) to good (ARI = 0.86), where $t = 100$ and 500 seemed to be sufficient to produce decent recovery for the GMM and *k*-means, respectively.

The accuracy of cluster membership assignment changed unsteadily with varying proportions of the VAR(2). In the case when all the ILDs were generated as VAR(2) processes, the VAR(1)-based clustering (i.e., the LO) yielded a better recovery (ARI = 0.87) than the VAR(2)-based clustering (i.e., HO; ARI = 0.65). When all the ILDs followed a VAR(1), clustering based on a VAR(2) was associated with poor membership recovery, ARI = 0.38 for the GMM and ARI = 0.50 for the *k*-means. When they were clustering based on a VAR(1), the accuracy on average was decent, 0.85 for the GMM, and 0.74 for the *k*-means. Overall, the results suggested that underfitting is generally preferable to overfitting for recovering cluster membership with the VAR-based approach.

The recovery on cluster-specific dynamics was assessed by computing the mean Euclidean distance between the estimated and the true cluster-specific dynamic coefficients in the simulated data sets where the number of clusters was correctly recovered. The cluster-specific dynamic coefficients, including both AR and CR coefficients, were obtained by averaging the person-specific coefficients estimated within the cluster. A smaller Euclidean distance indicated a better estimation of the cluster-specific dynamics. Given that clustering algorithms always use arbitrary numbers to label clusters, we permuted the partitions before calculating the Euclidean distance to maximize the overlap between the estimated and the true clusters.

Table 3 shows that unlike those observed for the recovery of the number of clusters and cluster membership, the HO consistently exhibited superior recovery of cluster-specific coefficients than the LO under all the conditions where the VAR(2) processes were present, with the mean Euclidean distances of 0.025 and 0.063 for the HO and LO, respectively. In addition, the performance of the HO was noticeably improved with increasing sample size, greater length of the series, larger between-cluster distance, and greater proportion of effective coefficients. Regarding the clustering algorithms, it was evident the *k*-means performed comparably or even slightly better than the GMM when the VAR(2) were present.

Under the LO, the recovery of cluster-specific dynamic coefficients slightly decreased as the between-cluster distances and the number of effective coefficients increased. This became more pronounced when the proportion of ILDs following VAR(2) increased. However, this pattern was not observed under the conditions where all the ILDs were generated by VAR(1). Indeed, when these ILDs were clustered based on their true generative VAR(1) model, the cluster-specific dynamics were recovered extremely well, with the mean difference between the recovered and the true being as small as 0.015. All this indicated that when the Lag 2 processes were present, the Lag 2 coefficients became more important to the clustering as the between-cluster distances and the number of effective coefficients increased at both lags, under which underfitting with the LO strategy (ignoring the Lag 2 effect) would lead to poorer estimates of the cluster-specific coefficients.

So far, our simulation has shown that the LO was preferable for cluster and membership identification, whereas the HO outperformed on recovery of cluster-specific coefficients. However, in real-world applications, it is not legitimate to use the LO and HO as a sequence of steps for VAR-based clustering because they generally produce different clusters and then yield different coefficient estimates for the identified clusters. We proposed that the best solution would be to first apply the LO for cluster and membership identification, and then fit the HO VAR model to individual ILDs within each cluster for a more accurate person- and cluster-specific coefficient estimation. We added additional simulation analyses to test this proposed method. The results in Table 3 shows that after the clusters were identified using the LO, fitting a VAR(2) to each individual ILDs produced better cluster-specific estimates than those obtained by the HO. This held true for both the GMM and the *k*-means, with a higher improvement observed for the GMM.

To enhance the generalizability of our findings, we conducted three additional sets of simulation analyses with (a) a new set of parameter values for the AR and CR coefficients, (b) a new set of parameter values for the between-individual variations on the AR and CR coefficients within clusters, and (c) normal distributions instead of uniform distributions for generating the AR and CR coefficients, respectively. The results, together with a more detailed description of the additional analyses, can be found in the online supplemental materials. As shown, all the findings from the primary simulation study were successfully replicated.

Table 2
The Recovery of the Cluster Membership (ARI)

	Clustering algorithms							
	GMM				*k*-means			
	Temporal order selection methods							
	LO		HO		LO		HO	
Level	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
Proportion of VAR(2)								
0%	0.85	0.32	0.38	0.45	0.74	0.4	0.5	0.48
10%	0.87	0.29	0.52	0.45	0.59	0.42	0.39	0.44
40%	0.8	0.25	0.51	0.41	0.71	0.31	0.54	0.41
60%	0.93	0.19	0.66	0.44	0.85	0.29	0.67	0.44
90%	0.9	0.24	0.64	0.43	0.65	0.4	0.49	0.45
100%	0.91	0.24	0.63	0.45	0.83	0.32	0.67	0.45
Number of true clusters								
2	0.9	0.26	0.56	0.47	0.63	0.41	0.47	0.45
3	0.9	0.24	0.57	0.44	0.81	0.33	0.61	0.46
4	0.83	0.28	0.54	0.43	0.75	0.35	0.56	0.45
Sample size								
60	0.8	0.32	0.44	0.45	0.72	0.38	0.54	0.46
120	0.89	0.24	0.45	0.45	0.73	0.37	0.55	0.46
240	0.94	0.18	0.79	0.35	0.73	0.37	0.55	0.45
Length of ILDs								
50	0.69	0.36	0.24	0.35	0.42	0.39	0.09	0.24
100	0.87	0.26	0.4	0.43	0.68	0.37	0.36	0.42
500	0.97	0.12	0.74	0.4	0.89	0.25	0.84	0.3
1,000	0.97	0.12	0.85	0.32	0.92	0.22	0.89	0.25
Between-cluster distance								
0.1	0.77	0.35	0.4	0.44	0.56	0.43	0.41	0.45
0.15	0.87	0.26	0.52	0.45	0.7	0.38	0.5	0.47
0.2	0.92	0.2	0.62	0.44	0.8	0.32	0.59	0.45
0.25	0.95	0.16	0.69	0.41	0.86	0.27	0.68	0.41
Proportion of effective coefficients								
25%	0.8	0.33	0.42	0.44	0.63	0.41	0.45	0.46
37.50%	0.86	0.28	0.51	0.45	0.7	0.39	0.51	0.46
50%	0.89	0.25	0.57	0.45	0.74	0.36	0.55	0.46
62.50%	0.91	0.22	0.62	0.44	0.77	0.34	0.59	0.45
75%	0.92	0.2	0.66	0.43	0.8	0.33	0.63	0.44

Note. ARI = Adjusted Rand Index; GMM = Gaussian mixture model; LO = lowest order; HO = highest order; VAR = vector autoregressive; ILD = intensive longitudinal data.

Empirical Analysis

To further illustrate the uses of the VAR-based clustering techniques, we conducted an empirical analysis using the data collected from the "How Nuts are the Dutch" project (Van der Krieke et al., 2016). Data access can be requested by contacting Bertus Jeronimus. The original data set comprised 1,393 participants from the general population of the Netherlands who were assessed three times per day (morning, midday, and night) over 30 days. The variables used here were momentary affect measured by 12 items based on the circumplex model of affect (Barrett & Russell, 1999; Yik et al., 1999). Positive affect (PA) was assessed by enthusiastic, energetic, cheerful, relaxed, content, and calm, and negative affect (NA) was assessed by irritable, nervous, anxious, tired, dull, and gloomy. Each item was scored on a continuous scale ranging from 0 to 100. The values of NA and PA were computed as the averages of their respective item scores. We included the participants who completed at least 90% of the assessments, which resulted in a sample of 227 participants with $t \geq 81$ for the analysis.

To meet the assumption of equidistant time points required by VAR models, we added one measurement as a missing value between the previous night and the next morning for each participant. We then imputed the missing values using a bootstrapping-based algorithm built in the Amelia II program (Honaker et al., 2011). To enable the partitioning of individuals based solely on the dynamic patterns, the levels of the ILDs were removed by centering PA and NA around their respective means of each participant. R code for this empirical analysis can be found at https://github.com/ild-clustering/temporal_order_selection.

Results

We first fitted person-specific VAR models and determined the best temporal orders for each individual's NA and PA processes. It turned out that 144 of them were best described by a VAR(1) model, 46 by a VAR(2) model, and 37 by a VAR(3) model. In Step 1 of the clustering procedure, we fitted a VAR(1) to all individuals' affect processes (i.e., the LO strategy) and then a VAR(3) to all processes (i.e., the HO strategy). The estimated, person-specific

Table 3
The Differences Between the Estimated and the True Cluster-Specific Coefficients

	Clustering algorithms											
	GMM						*k*-means					
	Temporal order selection methods											
	LO		HO		LO with VAR(2)[a]		LO		HO		LO with VAR(2)[a]	
Level	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
Proportion of VAR(2)												
0%	0.016	0.012	0.028	0.026	0.02	0.014	0.015	0.01	0.019	0.011	0.018	0.012
10%	0.042	0.013	0.032	0.026	0.024	0.016	0.04	0.011	0.024	0.015	0.022	0.014
40%	0.047	0.013	0.029	0.024	0.022	0.014	0.052	0.014	0.026	0.015	0.024	0.014
60%	0.072	0.022	0.023	0.019	0.02	0.014	0.074	0.023	0.019	0.012	0.02	0.013
90%	0.077	0.022	0.029	0.022	0.024	0.016	0.083	0.022	0.027	0.018	0.024	0.015
100%	0.113	0.025	0.022	0.018	0.019	0.013	0.115	0.025	0.017	0.011	0.019	0.013
Number of true clusters												
2	0.056	0.031	0.026	0.02	0.02	0.015	0.058	0.037	0.023	0.015	0.018	0.012
3	0.062	0.037	0.027	0.024	0.021	0.014	0.064	0.039	0.021	0.013	0.021	0.013
4	0.068	0.043	0.029	0.023	0.024	0.015	0.073	0.044	0.021	0.013	0.024	0.015
Sample size												
60	0.066	0.038	0.042	0.033	0.028	0.018	0.067	0.04	0.025	0.014	0.026	0.016
120	0.061	0.037	0.022	0.019	0.02	0.013	0.065	0.041	0.022	0.013	0.02	0.013
240	0.059	0.037	0.017	0.012	0.016	0.011	0.063	0.041	0.018	0.013	0.017	0.011
Length of ILDs												
50	0.075	0.033	0.068	0.032	0.044	0.012	0.085	0.037	0.075	0.028	0.048	0.013
100	0.063	0.036	0.036	0.018	0.026	0.008	0.068	0.039	0.037	0.013	0.027	0.008
500	0.055	0.038	0.011	0.005	0.01	0.003	0.057	0.04	0.01	0.003	0.01	0.003
1,000	0.054	0.039	0.007	0.003	0.007	0.002	0.056	0.04	0.007	0.002	0.007	0.002
Between-cluster distance												
0.1	0.058	0.03	0.03	0.025	0.022	0.014	0.062	0.033	0.021	0.012	0.021	0.012
0.15	0.06	0.034	0.028	0.023	0.022	0.014	0.064	0.037	0.021	0.013	0.021	0.013
0.2	0.062	0.038	0.027	0.022	0.021	0.014	0.065	0.041	0.022	0.013	0.021	0.014
0.25	0.067	0.044	0.025	0.02	0.021	0.015	0.069	0.046	0.023	0.015	0.021	0.014
Proportion of effective coefficients												
25%	0.059	0.032	0.029	0.026	0.022	0.014	0.062	0.036	0.02	0.012	0.022	0.013
37.50%	0.061	0.035	0.028	0.023	0.022	0.014	0.064	0.039	0.023	0.014	0.022	0.014
50%	0.062	0.037	0.026	0.021	0.022	0.014	0.065	0.04	0.022	0.014	0.021	0.014
62.50%	0.063	0.04	0.026	0.021	0.021	0.015	0.066	0.043	0.022	0.014	0.021	0.014
75%	0.064	0.041	0.026	0.02	0.021	0.014	0.067	0.044	0.022	0.013	0.021	0.014

Note. GMM = Gaussian mixture model; LO = lowest order; HO = highest order; VAR = vector autoregressive; ILD = intensive longitudinal data.
[a] LO was used to identify individual membership. Cluster-specific coefficients calculated by averaging person-specific dynamic coefficients which based on a VAR(2) model across the individual.

dynamic coefficients were then fed to the GMM and *k*-means algorithms for clustering in Step 2. The optimal number of clusters was determined by using BIC under the GMM and by the Gap statistics under the *k*-means algorithm. The BIC with GMM indicated a three-cluster solution under both the LO and HO strategies, whereas the Gap statistics with the *k*-means suggested a one-cluster solution. Given that the simulation study showed a superior performance of the GMM over the *k*-means on cluster identification, we determined that three clusters were optimal for the NA and PA processes observed in the current sample.

The individual ILDs were assigned to one and only one of the three clusters identified through the VAR(1)-based clustering in conjunction with the GMM algorithm, but the cluster-specific coefficients were computed by averaging the person-specific coefficients estimated by fitting a VAR(3) to each ILD. This decision was made based on our findings from the simulation study—when the ILDs followed a mixed order of dynamical processes, the LO performed better than the HO in cluster and membership identification, and after the clustering stage, fitting a higher-order model yielded more accurate estimation of the cluster-specific coefficients.

To better understand the nature of the identified clusters, we examined the associations between the clusters and the Big-Five personality traits (NEO-FFI-3; de Fruyt & Hoekstra, 2014), happiness (Fordyce, 1988), and the status of depression, anxiety, and stress scale (Lovibond & Lovibond, 1995). We also included relevant demographic variables such as age and gender to aid the description of each cluster in Table 4.

Cluster 1 contained most individuals (64.3%) with 79% of them being females. The members in this cluster had a moderate level of mean PA (54.9) and NA (18.3) with the PA and NA correlated at −0.57, $p < .001$ (see Table 4). Table 5 shows that this cluster of people displayed relatively strong AR effects for both PA (0.44) and NA (0.40) at Lag 1; however, they showed practically zero CR relationship between PA and NA, suggesting that PA and NA were independently unfolding across time for the people in this cluster. Individuals in this cluster showed a weak negative AR effect for PA (−0.10) at Lag 2, suggesting that their PA tended to return to the

Table 4
Observed Features Within Each Identified Cluster

Cluster	Proportion	Age (*SD*)	Female (%)	PA (*SD*)	NA (*SD*)	Correlation of PA and NA
1	64.3% (146)	41.9 (0.21)	78.7	54.9 (15.5)	18.3 (16.8)	−.57 ($p < .001$)
2	29.5% (67)	38.0 (0.06)	93.6	52.5 (17.1)	21.7 (17.1)	−.58 ($p < .001$)
3	6.2% (14)	48.4 (0.24)	75.9	60.8 (15.0)	7.86 (9.03)	−.23 ($p = .43$)

Note. PA = positive affect; NA = negative affect.

cluster's average over time. As shown in Table 6, this cluster was associated with moderate levels of depression, anxiety, stress, and happiness, in comparison with the other two clusters.

Cluster 2 included 29.5% of the participants with 93.6% being females. People in this cluster had a low mean PA (52.5) and a high mean NA (21.7) with the PA and NA correlated at −0.58, $p < .001$ (see Table 4). In comparison with other clusters, this cluster showed the weakest AR effect on PA (0.34) but the strongest on NA (0.57) at Lag 1, indicating that these people experienced long-lasting NA and short-lasting PA relative to members in other clusters. Meanwhile, people in this cluster also scored the highest on depression, anxiety, and stress and the lowest on happiness. Unlike the zero CR relationship between PA and NA observed in Cluster 1, people in Cluster 2 showed a nontrivial Lag 1 effect (−0.15) from NA ($t - 1$) to PA (t), indicating that the NA experienced earlier of the day tended to decrease their PA later of the day. Like the members in Cluster 1, people in this cluster showed a weak negative AR effect for PA (−0.13) at Lag 2, suggesting that their PA tended to return to the cluster's average over time. Regarding their personality profiles, people in Cluster 2 scored the highest on neuroticism among the three clusters (Figure 1). Putting it all together, the result was consistent with previous literature that neuroticism was associated with an increased likelihood of a NA at one moment being carried over to the next moment (e.g., Suls et al., 1998).

Cluster 3 captured 6.2% of the sample with only 14 participants. The members in this cluster had a high mean PA (60.8) and a low mean NA (7.86) with the PA and NA correlated at −0.23, $p = 0.43$, as shown in Table 4. People in this group showed the opposite pattern of AR effect to those observed in Cluster 2; namely, they had the strongest Lag 1 AR effects on PA (0.52) but the lowest on NA (0.19) among the three clusters. In other words, they experienced long-lasting PA and short-lasting NA relative to people in other clusters. Members in this cluster also showed a strong Lag 1 effect (0.37) from NA ($t - 1$) to PA (t), suggesting that the NA they experience at one moment tended to trigger a PA at the next moment. At Lag 2, the PA demonstrated a weak negative AR effect (−0.13), suggesting the PA tended to return to the cluster's average over time. In addition, NA showed a Lag-2 effect on PA (−0.13), indicating that the current NA would attenuate the PA two units later in time. Members in this cluster scored the highest on happiness and the lowest on depression, stress, and anxiety. They also scored the highest on conscientiousness and the lowest on neuroticism that has been shown to be associated with high levels of well-being (e.g., Soto, 2015).

Table 5
The Estimated Cluster-Specific AR and CR Coefficients for Each Identified Cluster

Cluster	PA_{t-1}	NA_{t-1}	PA_{t-2}	NA_{t-2}	PA_{t-3}	NA_{t-3}
Cluster 1						
PA_t	0.438	−0.004	−0.102	−0.029	0.073	−0.011
NA_t	−0.106	0.397	0.011	−0.044	0.008	0.066
Cluster 2						
PA_t	0.341	−0.150	−0.129	−0.049	0.070	0.006
NA_t	−0.001	0.572	0.032	−0.044	0.001	0.091
Cluster 3						
PA_t	0.523	0.370	−0.130	−0.125	0.077	−0.072
NA_t	−0.081	0.190	0.040	0.010	0.032	0.028

Note. AR = autoregressive; CR = cross-regressive; PA = positive affect; NA = negative affect.

Discussion

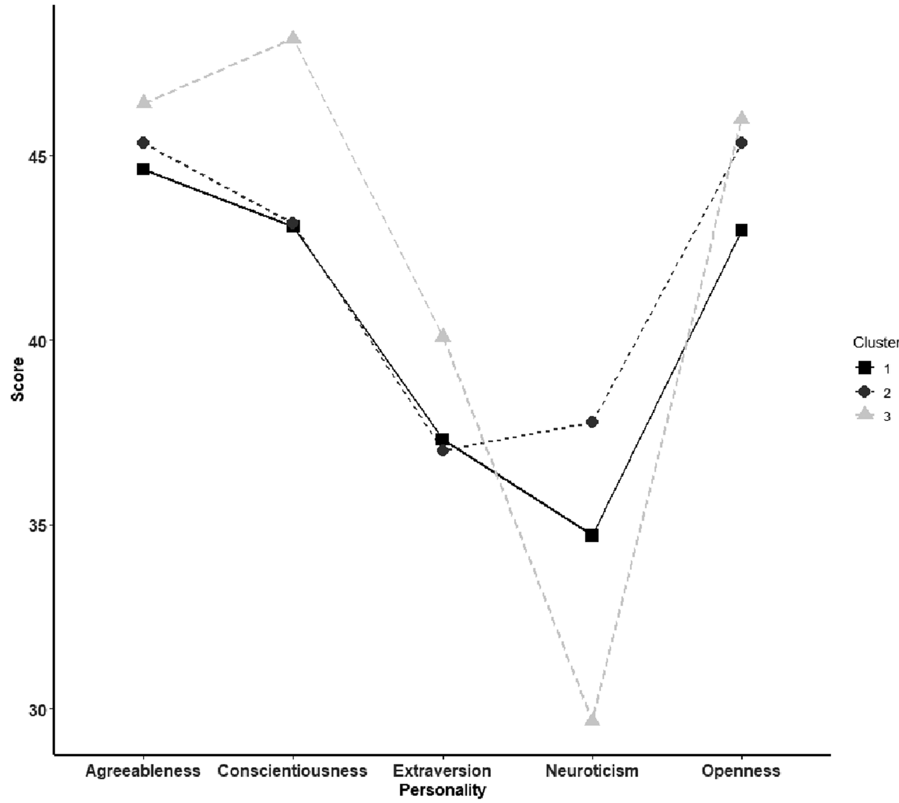
Model-based clustering of ILDs helps identify unobservable groups based on the dynamical features of the intensively collected data. When this technique is implemented, the temporal order of the studied processes needs to be set identical using either the HO or LO strategies. Such practice likely leads to model overfitting or underfitting, which in turn could impact the clustering outcomes. This study aimed to examine this imperative but underaddressed issue by evaluating the performance of the HO and LO in conjunction with two different clustering algorithms, in the hope to provide insight and guidance on the uses of the model-based clustering techniques.

Our simulation analyses revealed that in general the LO yielded superior performance over the HO on cluster identification, whereas the HO outperformed the LO on recovering cluster-specific dynamics. In other words, underfitting of the VAR models was found generally preferable over overfitting when the goal is to identify clusters and assign individuals to the correct clusters; however, overfitting would be preferable for estimating dynamic coefficients to characterize the identified clusters. It is well understood that fitting a high-order VAR model means to estimate a large number of dynamic coefficients, resulting in the feature space to be too large for the clustering algorithm to work on. When the HO is used together with a high-order VAR, the feature space is not only too large but also likely to contain a substantial amount of noninformative features including zero values, which would attenuate the clustering accuracy to a greater extent for both the *k*-means and GMM algorithms (e.g., Davis et al., 2016; Ruan et al., 2011; Steinley & Brusco, 2008;

Table 6
Mean (Standard Deviation) of the External Variables Within the Identified Clusters

Cluster	Depression (0–42)	Anxiety (0–42)	Stress (0–42)	Happiness (0–10)
1	7.18 (7.33)	3.64 (4.53)	8.88 (6.75)	6.72 (1.45)
2	8.34 (7.73)	4.70 (5.05)	11.1 (7.52)	6.47 (1.41)
3	2.31 (3.07)	0.85 (0.99)	5.92 (3.93)	7.86 (0.86)

Figure 1
Big-Five Personality Trait Profile for Each Identified Cluster



Takano et al., 2020). Note that the LO is not a perfect solution either because the correlated residuals yielded among the ILDs could negatively impact the clustering outcomes. However, our results showed that the implementation of HO attenuated the clustering accuracy much more than the LO did. This could also explain the fact that research on clustering multivariate ILDs has mostly been conducted based on first-order models (e.g., Bulteel et al., 2016; Ernst et al., 2021). Such practices not only ease the implementation of the clustering techniques but also likely yield higher clustering accuracy.

One piece of the findings deserves a special attention and more explanation; that is, when all the ILDs were generated by a VAR(2) model, the LO (i.e., fitting a VAR(1), the misspecified model) still outperformed the HO (i.e., fitting a VAR(2), the true model) on identifying correct clusters, while the former did worse than the latter in recovering cluster-specific coefficients. Regarding the recovery of true clusters, this result looks counterintuitive on the surface, as people often expect the true model would perform better than a seemingly wrong model. However, this makes legitimate sense when we look closely at possible unalignment between the performance of clustering and parameter estimation. While estimation accuracy depends on the correctness of the fitted model, clustering accuracy depends on the similarity or distance between individual estimates within a cluster and the dissimilarity across clusters. Thus, it is understandable that unbiased estimates do not necessarily translate into superior performance of clustering.

Furthermore, research has shown that in finite samples, underfitting leads to biased estimation but with small variances, whereas overfitting yields unbiased estimation but with large variances (e.g., de Waele & Broersen, 2003; Yarkoni & Westfall, 2017). At Step 1 of the VAR-based clustering procedure, fitting a VAR(1) to the ILDs of a VAR(2) leads to underfitting, which would yield biased estimates of the person-specific dynamic coefficients; however, the variance of the estimation tends to be small. In other words, the estimated person-specific coefficients within a cluster remain close to each other, although they differ substantially from their true cluster-specific values, thereby increasing the likelihood of being partitioned into the same cluster. In contrast, fitting a VAR(2) to the ILDs of a VAR(2), despite yielding unbiased estimates, the variance of the estimation tends to be greater than those obtained from underfitting. In this case, the estimated person-specific coefficients could deviate substantially from their cluster-specific parameter values, which would augment the clustering inaccuracy.

In terms of estimating cluster-specific dynamics, the VAR-based clustering with the HO strategy tended to outperform the LO strategy when the ILDs were mixed with dynamical processes of different orders. These findings were consistent with the previous research showing that an overfitted model enables to yield unbiased estimation (D. M. McNeish, 2015; Yarkoni & Westfall, 2017), although overfitting of time series models with high temporal orders severely restricted clustering accuracy (Bouveyron & Brunet-Saumard, 2014; Scott & Thompson, 1983). However, in real-world applications, it is not legitimate to use the LO and HO as a sequence of steps for VAR-based clustering. As demonstrated by our additional simulation analyses, the best solution seemed to first apply the LO for cluster identification and membership assignment, and then fit the HO VAR model to individual ILDs within each cluster for a more accurate person- and cluster-specific coefficient estimation.

In addition to the LO and HO, we also investigated the performance of the GMM and k-means algorithms. Overall, the GMM demonstrated higher accuracy of recovery on the number of clusters and the cluster membership; however, both algorithms performed comparably well on the recovery of cluster-specific dynamics. It has been evidenced that the GMM as a probabilistic algorithm performs well in identifying clusters when the coefficients of a VAR model come from a mixture of multivariate normal distributions (Banfield & Raftery, 1993; Browne & McNicholas, 2014; Celeux & Govaert, 1995). As shown in our study, however, when the HO was applied, the GMM showed slightly lower recovery accuracy than the k-means while both algorithms performed poorly. It was evident again that when the feature space was large, clustering accuracy was attenuated even more for GMM than it was for k-means.

A special note should be made regarding sample size, length of ILDs, and the number of variables consisting of the dynamic system under study. First, short ILDs such as $t = 50$ were in general associated with unacceptably poor clustering outcomes, and $t = 100$ seemed to be a minimum requirement for VAR model-based clustering of ILDs. Second, increases of the sample size improved the accuracy of identification on the number of clusters and the cluster membership, with over 80% accuracy on both criteria at $n = 120$. Third, although large values of n and t made appreciable contributions to accuracy improvement, the performance of the GMM was more sensitive to n whereas the performance of the k-means was more sensitive to t. Although technical advancements have made the collection of ILDs much easier than ever, current research in social and behavioral sciences is still characterized by relatively small n and t. To obtain optimal clustering accuracy, methodologists and applied researchers could consider adopting the techniques for estimating high-order models with small t (e.g., Basu & Michailidis, 2015; Brodinová et al., 2019; Hsu et al., 2008; Raftery & Dean, 2006; S. Song & Bickel, 2011), as well as those for clustering ILDs with small n (e.g., Levina et al., 2008; Maugis et al., 2009). Finally, when the number of variables of being modeled is large, variable selection techniques could be used to reduce the number of estimated coefficients, thereby reducing the size of the resulting feature space to improve the clustering outcomes (e.g., Bouveyron & Brunet-Saumard, 2014; Nicholson et al., 2020). In addition, a weighting procedure proposed by Steinley and Brusco (2008) could be implemented in k-means clustering to unmask cluster structure by mitigating noninformative coefficients. Future studies are needed to explore more on the performance of such applications.

In the past, the model-based approach has been successfully applied to clustering univariate ILDs with AR (1) models, where the first-order AR coefficients are the only coefficients used for cluster identification. When it comes to clustering multivariate ILDs with VAR ($p > 1$), all the estimated AR and CR coefficients at the same or different lags are fed to a specific clustering algorithm and treated equally important in the clustering stage. This may contribute to the clustering inaccuracy particularly when the AR coefficients are more informative than the CR coefficients, and/or the coefficients at lower lags are more important than those at higher lags. Researchers have proposed using the cepstral distance to denote the position of the AR coefficients, through which the AR coefficients at lower lags are deemed as more important than those at higher lags in the clustering process (Kalpakis et al., 2001). This method was developed only for univariate ILDs and has not yet been available for multivariate ILDs.

The model-based clustering requires the number of features (or model coefficients) used for clustering must be identical across all individuals. To reach this goal, one needs to use either the LO or the HO to set the lag order to be the same, during which the individual differences on the lag order were ignored. One possible way to take account of such differences is to adopt a two-stage approach, wherein individuals are precategorized into subgroups based on the lag order and then the model-based clustering is undertaken within each subgroup. However, this approach is not feasible for typical behavioral research. First, the subgroup sizes are likely to vary substantially with some subgroups being very small, and small sample size detrimentally affected the clustering accuracy, as shown in our simulation study. For example, in our empirical analysis, only 37 out of 227 individuals followed a VAR(3). If a clustering based on a VAR(3) model is performed with 37 ILDs, the accuracy of the clustering will not be trustworthy. Second, the lag order selection itself may be inaccurate, and therefore, the clustering accuracy could be further undermined. Research has shown that AIC and BIC, as standard information criteria for determining the lag order, are more likely to underestimate the true lag orders in shorter ILDs (Kilian, 2001). Future research is needed to study the feasibility of this approach for the ILDs of large sizes with long series.

A few limitations need to be noted about our study. First, the number of ILD variables was set to be four in all simulated conditions, which to some degree restricts the generalizability of our findings to the VAR-based clustering with much fewer and many more ILD variables. Given that dynamic analyses in social and behavioral research often involve more than four variables (e.g., Bosley et al., 2020; Bringmann et al., 2013, 2016; Mansueto et al., 2023; Pe et al., 2015; Snippe et al., 2015; Zheng et al., 2013), future research is needed to study the clustering techniques with large numbers of ILD variables. Second, in our study, all the data were simulated to be stationary as required by the VAR models. Thus, our findings may not be applicable when ILDs are not stationary such as displaying linear trends over time. Although preprocessing procedures (e.g., differencing and detrending) can be used to render the ILDs stationary before clustering, their potential impact on the accuracy of VAR-based clustering is unknown. As a result, we recommend researchers to cautiously apply the VAR-based clustering techniques when stationarity is not met for the ILDs under study. Third, our findings from the simulation study were based on the data generated with the VAR(2) as the HO model. Although we expect these findings to be applicable to conditions with the highest p greater than 2, our current study did not offer a direct examination on this. It is likely that the choice of HO versus LO strategies would be more complicated in those conditions.

In summary, evidence from our simulation study stated that when the VAR-based approach was used to clustering multivariate ILDs with mixing orders of processes: (a) the LO generally outperformed the HO in identification of clusters, (b) the HO was more favorable than the LO in estimation of cluster-specific dynamics, (c) the GMM generally outperformed the k-means, and (d) the LO in conjunction with the GMM produced the best cluster identification outcome. Our empirical analysis also demonstrated that the uses of the LO combined with the GMM yielded meaningful and interpretable results from clustering multivariate affect data. Such analysis indeed offered an innovative perspective to look at the dynamic nature of momentary affect within each distinct group and its relation to personality traits and well-being. Putting it all together, we would recommend applied researchers to identify clusters using the VAR-based approach along with the LO in conjunction with the GMM but to estimate the cluster-specific dynamics by fitting a higher-order VAR model to individual ILDs within clusters. Future research is much needed to further deepen our understanding and aid the applications of these techniques particularly in social and behavioral research where *n* and *t* are typically small.

References

Aghabozorgi, S., Shirkhorshidi, A. S., & Wah, T. Y. (2015). Time-series clustering—A decade review. *Information Systems*, *53*, 16–38. https://doi.org/10.1016/j.is.2015.04.007

Aghabozorgi, S., & Teh, Y. W. (2014). Stock market co-movement assessment using a three-phase clustering method. *Expert Systems with Applications*, *41*(4), 1301–1314. https://doi.org/10.1016/j.eswa.2013.08.028

Asparouhov, T., Hamaker, E. L., & Muthén, B. (2018). Dynamic structural equation models. *Structural Equation Modeling: A Multidisciplinary Journal*, *25*(3), 359–388. https://doi.org/10.1080/10705511.2017.1406803

Axelson, D. A., Bertocci, M. A., Lewin, D. S., Trubnick, L. S., Birmaher, B., Williamson, D. E., Ryan, N. D., & Dahl, R. E. (2003). Measuring mood and complex behavior in natural environments: Use of ecological momentary assessment in pediatric affective disorders. *Journal of Child and Adolescent Psychopharmacology*, *13*(3), 253–266. https://doi.org/10.1089/104454603322572589

Babbin, S. F., Velicer, W. F., Aloia, M. S., & Kushida, C. A. (2015). Identifying longitudinal patterns for individuals and subgroups: An example with adherence to treatment for obstructive sleep apnea. *Multivariate Behavioral Research*, *50*(1), 91–108. https://doi.org/10.1080/00273171.2014.958211

Banfield, J. D., & Raftery, A. E. (1993). Model-based Gaussian and non-Gaussian clustering. *Biometrics*, *49*(3), 803–821. https://doi.org/10.2307/2532201

Barrett, L. F., & Russell, J. A. (1999). The structure of current affect: Controversies and emerging consensus. *Current Directions in Psychological Science*, *8*(1), 10–14. https://doi.org/10.1111/1467-8721.00003

Basu, S., & Michailidis, G. (2015). Regularized estimation in sparse high-dimensional time series models. *The Annals of Statistics*, *43*(4), 1535–1567. https://doi.org/10.1214/15-AOS1315

Bosley, H. G., Sandel, D. B., & Fisher, A. J. (2020). Idiographic dynamics of positive affect in GAD: Modeling emotion regulation at the person level. *European Journal of Psychological Assessment*, *36*(3), 500–509. https://doi.org/10.1027/1015-5759/a000580

Bouveyron, C., & Brunet-Saumard, C. (2014). Model-based clustering of high-dimensional data: A review. *Computational Statistics & Data Analysis*, *71*, 52–78. https://doi.org/10.1016/j.csda.2012.12.008

Braun, P. A., & Mittnik, S. (1993). Misspecifications in vector autoregressions and their effects on impulse responses and variance decompositions. *Journal of Econometrics*, *59*(3), 319–341. https://doi.org/10.1016/0304-4076(93)90029-5

Bringmann, L. F., Pe, M. L., Vissers, N., Ceulemans, E., Borsboom, D., Vanpaemel, W., Tuerlinckx, F., & Kuppens, P. (2016). Assessing temporal emotion dynamics using networks. *Assessment*, *23*(4), 425–435. https://doi.org/10.1177/1073191116645909

Bringmann, L. F., Vissers, N., Wichers, M., Geschwind, N., Kuppens, P., Peeters, F., Borsboom, D., & Tuerlinckx, F. (2013). A network approach to psychopathology: New insights into clinical longitudinal data. *PLoS ONE*, *8*(4), Article e60188. https://doi.org/10.1371/journal.pone.0060188

Brodinová, Š., Filzmoser, P., Ortner, T., Breiteneder, C., & Rohm, M. (2019). Robust and sparse k-means clustering for high-dimensional data. *Advances in Data Analysis and Classification*, *13*(4), 905–932. https://doi.org/10.1007/s11634-019-00356-9d

Browne, R. P., & McNicholas, P. D. (2014). Estimating common principal components in high dimensions. *Advances in Data Analysis and Classification*, *8*(2), 217–226. https://doi.org/10.1007/s11634-013-0139-1

Bulteel, K., Tuerlinckx, F., Brose, A., & Ceulemans, E. (2016). Clustering vector autoregressive models: Capturing qualitative differences in within-person dynamics. *Frontiers in Psychology*, *7*, Article 1540. https://doi.org/10.3389/fpsyg.2016.01540

Celeux, G., & Govaert, G. (1995). Gaussian Parsimonious clustering models. *Pattern Recognition*, *28*(5), 781–793. https://doi.org/10.1016/0031-3203(94)00125-6

Chow, S.-M., Hamagani, F., & Nesselroade, J. R. (2007). Age differences in dynamical emotion-cognition linkages. *Psychology and Aging*, *22*(4), 765–780. https://doi.org/10.1037/0882-7974.22.4.765

daSilva, A. W., Huckins, J. F., Wang, W., Wang, R., Campbell, A. T., & Meyer, M. L. (2021). Daily perceived stress predicts less next day social interaction: Evidence from a naturalistic mobile sensing study. *Emotion*, *21*(8), 1760–1770. https://doi.org/10.1037/emo0000994

Davis, R. A., Zang, P., & Zheng, T. (2016). Sparse vector autoregressive modeling. *Journal of Computational and Graphical Statistics*, *25*(4), 1077–1096. https://doi.org/10.1080/10618600.2015.1092978

de Fruyt, F., & Hoekstra, H. (2014). *NEO-PI-3 persoonlijkheidsvragenlijst* [NEO-PI-3 Personality Questionnaires]. Hogrefe.

de Waele, S., & Broersen, P. M. T. (2003). Order selection for vector autoregressive models. *IEEE Transactions on Signal Processing*, *51*(2), 427–433. https://doi.org/10.1109/TSP.2002.806905

Elangasinghe, M. A., Singhal, N., Dirks, K. N., Salmond, J. A., & Samarasinghe, S. (2014). Complex time series analysis of PM10 and PM2. 5 for a coastal site using artificial neural network modelling and k-means clustering. *Atmospheric Environment*, *94*, 106–116. https://doi.org/10.1016/j.atmosenv.2014.04.051

Ernst, A. F., Albers, C. J., Jeronimus, B. F., & Timmerman, M. E. (2020). Inter-individual differences in multivariate time-series. *European Journal of Psychological Assessment*, *36*(3), 482–491. https://doi.org/10.1027/1015-5759/a000578

Ernst, A. F., Timmerman, M. E., Jeronimus, B. F., & Albers, C. J. (2021). Insight into individual differences in emotion dynamics with clustering. *Assessment*, *28*(4), 1186–1206. https://doi.org/10.1177/1073191119873714

Fordyce, M. W. (1988). A review of research on the happiness measures: A sixty second index of happiness and mental health. *Social Indicators Research*, *20*(4), 355–381. https://doi.org/10.1007/BF00302333

Fraley, C., & Raftery, A. E. (1998). How many clusters? Which clustering method? Answers via model-based cluster analysis. *The Computer Journal*, *41*(8), 578–588. https://doi.org/10.1093/comjnl/41.8.578

Fraley, C., & Raftery, A. E. (2002). Model-based clustering, discriminant analysis, and density estimation. *Journal of the American Statistical Association*, *97*(458), 611–631. https://doi.org/10.1198/016214502760047131

Gates, K. M., Lane, S. T., Varangis, E., Giovanello, K., & Guskiewicz, K. (2017). Unsupervised classification during time-series model building. *Multivariate Behavioral Research*, *52*(2), 129–148. https://doi.org/10.1080/00273171.2016.1256187

Gates, K. M., Molenaar, P. C., Hillary, F. G., Ram, N., & Rovine, M. J. (2010). Automatic search for fMRI connectivity mapping: An alternative to Granger causality testing using formal equivalences among SEM path modeling, VAR, and unified SEM. *NeuroImage*, *50*(3), 1118–1125. https://doi.org/10.1016/j.neuroimage.2009.12.117

Gu, F., Preacher, K. J., & Ferrer, E. (2014). A state space modeling approach to mediation analysis. *Journal of Educational and Behavioral Statistics*, *39*(2), 117–143. https://doi.org/10.3102/1076998614524823

Hamaker, E. L., Asparouhov, T., Brose, A., Schmiedek, F., & Muthén, B. (2018). At the frontiers of modeling intensive longitudinal data:

Dynamic structural equation models for the affective measurements from the COGITO study. *Multivariate Behavioral Research*, *53*(6), 820–841. https://doi.org/10.1080/00273171.2018.1446819

Hamilton, J. D. (1994). *Time series analysis*. Princeton University Press.

Honaker, J., King, G., & Blackwell, M. (2011). Amelia II: A program for missing data. *Journal of Statistical Software*, *45*(7), 1–47. https://doi.org/10.18637/jss.v045.i07

Hsu, N. J., Hung, H. L., & Chang, Y. M. (2008). Subset selection for vector autoregressive processes using Lasso. *Computational Statistics & Data Analysis*, *52*(7), 3645–3657. https://doi.org/10.1016/j.csda.2007.12.004

Hubert, L., & Arabie, P. (1985). Comparing partitions. *Journal of Classification*, *2*(1), 193–218. https://doi.org/10.1007/BF01908075

Jahng, S., Wood, P. K., & Trull, T. J. (2008). Analysis of affective instability in ecological momentary assessment: Indices using successive difference and group comparison via multilevel modeling. *Psychological Methods*, *13*(4), 354–375. https://doi.org/10.1037/a0014173

Kalpakis, K., Gada, D., & Puttagunta, V. (2001). *Distance measures for effective clustering of ARIMA time-series*. Proceedings 2001 IEEE International Conference on Data Mining (pp. 273–280). IEEE.

Kilian, L. (2001). Impulse response analysis in vector autoregressions with unknown lag order. *Journal of Forecasting*, *20*(3), 161–179. https://doi.org/10.1002/1099-131X(200104)20:3<161::AID-FOR770>3.0.CO;2-X

Kim, J., Zhu, W., Chang, L., Bentler, P. M., & Ernst, T. (2007). Unified structural equation modeling approach for the analysis of multisubject, multivariate functional MRI data. *Human Brain Mapping*, *28*(2), 85–93. https://doi.org/10.1002/hbm.20259

Krone, T., Albers, C. J., Kuppens, P., & Timmerman, M. E. (2018). A multivariate statistical model for emotion dynamics. *Emotion*, *18*(5), 739–754. https://doi.org/10.1037/emo0000384

Lafit, G., Meers, K., & Ceulemans, E. (2022). A systematic study into the factors that affect the predictive accuracy of multilevel VAR (1) models. *Psychometrika*, *87*(2), 432–476. https://doi.org/10.1007/s11336-021-09803-z

Levina, E., Rothman, A., & Zhu, J. (2008). Sparse estimation of large covariance matrices via a nested lasso penalty. *The Annals of Applied Statistics*, *2*(1), 245–263. https://doi.org/10.1214/07-AOAS139

Liao, T. W. (2005). Clustering of time series data—A survey. *Pattern Recognition*, *38*(11), 1857–1874. https://doi.org/10.1016/j.patcog.2005.01.025

Liu, S., Rovine, M. J., Klein, L. C., & Almeida, D. M. (2013). Synchrony of diurnal cortisol pattern in couples. *Journal of Family Psychology*, *27*(4), 579–588. https://doi.org/10.1037/a0033735

Lovibond, P. F., & Lovibond, S. H. (1995). The structure of negative emotional states: Comparison of the Depression Anxiety Stress Scales (DASS) with the Beck Depression and Anxiety Inventories. *Behaviour Research and Therapy*, *33*(3), 335–343. https://doi.org/10.1016/0005-7967(94)00075-U

Lütkepohl, H. (2015). *New introduction to multiple time series analysis*. Springer.

MacQueen, J. (1967). Classification and analysis of multivariate observations. In L. M. Le Cam & J. Neyman (Eds.), *Proceedings of the 5th Berkeley Symposium on Mathematical Statistics and Probability* (Vol. 1, pp. 281–297). University of California Press.

Maharaj, E. A. (2000). Cluster of time series. *Journal of Classification*, *17*(2), 297–314. https://doi.org/10.1007/s003570000023

Mansueto, A. C., Wiers, R. W., van Weert, J. C. M., Schouten, B. C., & Epskamp, S. (2023). Investigating the feasibility of idiographic network models. *Psychological Methods*, *28*(5), 1052–1068. https://doi.org/10.1037/met0000466

Maugis, C., Celeux, G., & Martin-Magniette, M.-L. (2009). Variable selection for clustering with Gaussian mixture models. *Biometrics*, *65*(3), 701–709. https://doi.org/10.1111/j.1541-0420.2008.01160.x

McLachlan, G. J., & Basford, K. E. (1988). *Mixture models: Inference and applications to clustering*. Marcel Dekker.

McNeish, D., & Hamaker, E. L. (2020). A primer on two-level dynamic structural equation models for intensive longitudinal data in Mplus. *Psychological Methods*, *25*(5), 610–635. https://doi.org/10.1037/met0000250

McNeish, D. M. (2015). Using Lasso for predictor selection and to assuage overfitting: A method long overlooked in behavioral sciences. *Multivariate Behavioral Research*, *50*(5), 471–484. https://doi.org/10.1080/00273171.2015.1036965

Milligan, G. W., & Cooper, M. C. (1986). A study of the comparability of external criteria for hierarchical cluster analysis. *Multivariate Behavioral Research*, *21*(4), 441–458. https://doi.org/10.1207/s15327906mbr2104_5

Nicholson, W. B., Wilms, I., Bien, J., & Matteson, D. S. (2020). High dimensional forecasting via interpretable vector autoregression. *The Journal of Machine Learning Research*, *21*(1), 6690–6741.

Pe, M. L., Kircanski, K., Thompson, R. J., Bringmann, L. F., Tuerlinckx, F., Mestdagh, M., & Kuppens, P. (2015). Emotion-network density in major depressive disorder. *Clinical Psychological Science*, *3*(2), 292–300. https://doi.org/10.1177/2167702614540645

Pe, M. L., & Kuppens, P. (2012). The dynamic interplay between emotions in daily life: Augmentation, blunting, and the role of appraisal overlap. *Emotion*, *12*(6), 1320–1328. https://doi.org/10.1037/a0028262

Price, R. B., Lane, S., Gates, K., Kraynak, T. E., Horner, M. S., Thase, M. E., & Siegle, G. J. (2017). Parsing heterogeneity in the brain connectivity of depressed and healthy adults during positive mood. *Biological Psychiatry*, *81*(4), 347–357. https://doi.org/10.1016/j.biopsych.2016.06.023

Raftery, A. E., & Dean, N. (2006). Variable selection for model-based clustering. *Journal of the American Statistical Association*, *101*(473), 168–178. https://doi.org/10.1198/016214506000000113

Ren, B., & Barnett, I. (2022). Autoregressive mixture models for clustering time series. *Journal of Time Series Analysis*, *43*(6), 918–937. https://doi.org/10.1111/jtsa.12644

Roche, M. J., Pincus, A. L., Rebar, A. L., Conroy, D. E., & Ram, N. (2014). Enriching psychological assessment using a person-specific analysis of interpersonal processes in daily life. *Assessment*, *21*(5), 515–528. https://doi.org/10.1177/1073191114540320

Rosmalen, J. G., Wenting, A. M., Roest, A. M., de Jonge, P., & Bos, E. H. (2012). Revealing causal heterogeneity using time series analysis of ambulatory assessments: Application to the association between depression and physical activity after myocardial infarction. *Psychosomatic Medicine*, *74*(4), 377–386. https://doi.org/10.1097/PSY.0b013e3182545d47

Ruan, L., Yuan, M., & Zou, H. (2011). Regularized parameter estimation in high-dimensional Gaussian mixture models. *Neural Computation*, *23*(6), 1605–1622. https://doi.org/10.1162/NECO_a_00128

Scott, D. W., & Thompson, J. R. (1983). Probability density estimation in higher dimensions. In J. E. Gentle (Ed.), *Computer Science and Statistics: Proceedings of the Fifteenth Symposium on the Interface* (Vol. 528, pp. 173–179). North-Holland.

Scrucca, L., Fop, M., Murphy, T. B., & Raftery, A. E. (2016). Mclust 5: Clustering, classification and density estimation using Gaussian finite mixture models. *The R Journal*, *8*(1), 289–317. https://doi.org/10.32614/RJ-2016-021

Shiffman, S., Stone, A. A., & Hufford, M. R. (2008). Ecological momentary assessment. *Annual Review of Clinical Psychology*, *4*(1), 1–32. https://doi.org/10.1146/annurev.clinpsy.3.022806.091415

Snippe, E., Nyklíček, I., Schroevers, M. J., & Bos, E. H. (2015). The temporal order of change in daily mindfulness and affect during mindfulness-based stress reduction. *Journal of Counseling Psychology*, *62*(2), 106–114. https://doi.org/10.1037/cou0000057

Song, H., & Ferrer, E. (2012). Bayesian Estimation of random coefficient dynamic factor models. *Multivariate Behavioral Research*, *47*(1), 26–60. https://doi.org/10.1080/00273171.2012.640593

Song, H., & Zhang, Z. (2014). Analyzing multiple multivariate time series data using multilevel dynamic factor models. *Multivariate Behavioral Research*, *49*(1), 67–77. https://doi.org/10.1080/00273171.2013.851018

Song, S., & Bickel, P. J. (2011). *Large vector autoregressions*. arXiv. https://doi.org/10.48550/arXiv.1106.3915

Soto, C. J. (2015). Is happiness good for your personality? Concurrent and prospective relations of the big five with subjective well-being. *Journal of Personality*, *83*(1), 45–55. https://doi.org/10.1111/jopy.12081

Stefanovic, M., Rosenkranz, T., Ehring, T., Watkins, E. R., & Takano, K. (2022). Is a high association between repetitive negative thinking and negative affect predictive of depressive symptoms? A clustering approach for experience-sampling data. *Clinical Psychological Science*, *10*(1), 74–89. https://doi.org/10.1177/21677026211009495

Steinbach, M., Ertöz, L., & Kumar, V. (2004). The challenges of clustering high dimensional data. In *New directions in statistical physics: Econophysics, bioinformatics, and pattern recognition* (pp. 273–309). Springer. https://doi.org/10.1007/978-3-662-08968-2_16

Steinley, D. (2004). Properties of the Hubert-Arable Adjusted Rand Index. *Psychological Methods*, *9*(3), 386–396. https://doi.org/10.1037/1082-989X.9.3.386

Steinley, D., & Brusco, M. J. (2007). Initializing K-means batch clustering: A critical evaluation of several techniques. *Journal of Classification*, *24*(1), 99–121. https://doi.org/10.1007/s00357-007-0003-0

Steinley, D., & Brusco, M. J. (2008). A new variable weighting and selection procedure for K-means cluster analysis. *Multivariate Behavioral Research*, *43*(1), 77–108. https://doi.org/10.1080/00273170701836695

Steinley, D., & Brusco, M. J. (2011). Evaluating mixture modeling for clustering: Recommendations and cautions. *Psychological Methods*, *16*(1), 63–79. https://doi.org/10.1037/a0022673

Stetco, A., Zeng, X., & Keane, J. (2013). Fuzzy cluster analysis of financial time series and their volatility assessment. In *Proceedings of the 2013 IEEE International Conference on Systems, Man, and Cybernetics* (pp. 91–96). IEEE.

Suls, J., Green, P., & Hillis, S. (1998). Emotional reactivity to everyday problems, affective inertia, and neuroticism. *Personality and Social Psychology Bulletin*, *24*(2), 127–136. https://doi.org/10.1177/0146167298242002

Takano, K., Stefanovic, M., Rosenkranz, T., & Ehring, T. (2020). Clustering individuals on limited features of a vector autoregressive model. *Multivariate Behavioral Research*, *56*(5), 768–786. https://doi.org/10.1080/00273171.2020.1767532

Tibshirani, R., Walther, G., & Hastie, T. (2001). Estimating the number of clusters in a data set via the gap statistic. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *63*(2), 411–423. https://doi.org/10.1111/1467-9868.00293

Tran, D., & Wagner, M. (2002, February 3–6). Fuzzy c-means clustering-based speaker verification. In *Advances in Soft Computing—AFSS 2002: 2002 AFSS International Conference on Fuzzy Systems, Calcutta, India* (pp. 318–324). Springer.

van der Krieke, L., Jeronimus, B. F., Blaauw, F. J., Wanders, R. B. K., Emerencia, A. C., Schenk, H. M., Vos, S. D., Snippe, E., Wichers, M., Wigman, J. T., Bos, E. H., Wardenaar, K. J., & de Jonge, P. (2016). Hownutsarethedutch (HoeGekIsNL): A crowdsourcing study of mental symptoms and strengths. *International Journal of Methods in Psychiatric Research*, *25*(2), 123–144. https://doi.org/10.1002/mpr.1495

Von Leupoldt, A., Riedel, F., & Dahme, B. (2006). The impact of emotions on the perception of dyspnea in pediatric asthma. *Psychophysiology*, *43*(6), 641–644. https://doi.org/10.1111/j.1469-8986.2006.00453.x

Whalen, D. J., Silk, J. S., Semel, M., Forbes, E. E., Ryan, N. D., Axelson, D. A., Birmaher, B., & Dahl, R. E. (2007). Caffeine consumption, sleep, and affect in the natural environments of depressed youth and healthy controls. *Journal of Pediatric Psychology*, *33*(4), 358–367. https://doi.org/10.1093/jpepsy/jsm086

Xiong, Y., & Yeung, D. Y. (2004). Time series clustering with ARMA mixtures. *Pattern Recognition*, *37*(8), 1675–1689. https://doi.org/10.1016/j.patcog.2003.12.018

Yan, W., Zhao, M., Fu, Z., Pearlson, G. D., Sui, J., & Calhoun, V. D. (2022). Mapping relationships among schizophrenia, bipolar and schizoaffective disorders: A deep classification and clustering framework using fMRI time series. *Schizophrenia Research*, *245*, 141–150. https://doi.org/10.1016/j.schres.2021.02.007

Yarkoni, T., & Westfall, J. (2017). Choosing prediction over explanation in psychology: Lessons from machine learning. *Perspectives on Psychological Science*, *12*(6), 1100–1122. https://doi.org/10.1177/1745691617693393

Yik, M. S. M., Russell, J. A., & Barrett, L. F. (1999). Structure of self-reported current affect: Integration and beyond. *Journal of Personality and Social Psychology*, *77*(3), 600–619. https://doi.org/10.1037/0022-3514.77.3.600

Zheng, Y., Wiebe, R. P., Cleveland, H. H., Molenaar, P. C., & Harris, K. S. (2013). An idiographic examination of day-to-day patterns of substance use craving, negative affect, and tobacco use among young adults in recovery. *Multivariate Behavioral Research*, *48*(2), 241–266. https://doi.org/10.1080/00273171.2013.763012

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